

Coin Exchange:

Coin Exchange Algorithm is a greedy algorithm that finds the required amount using the minimum number of coins by selecting the largest possible coin first.

Algorithm

1. Start.
2. Read the coin denominations and amount.
3. Sort coins in descending order.
4. Select the largest coin.
5. Find the number of coins using:
 $\text{count} = \text{amount} // \text{coin}$
6. Subtract the selected coins from the amount:
 $\text{amount} = \text{amount} \% \text{coin}$
7. Repeat for all coins.
8. If the remaining amount is 0, the exact amount is formed.
9. Otherwise, exact amount cannot be formed.
10. Stop.

Out put:

```
Enter number of coins: 5
Enter coin 1: 1
Enter coin 2: 3
Enter coin 3: 5
Enter coin 4: 7
Enter coin 5: 9
Enter target amount: 30
Minimum number of coins: 4

...Program finished with exit code 0
Press ENTER to exit console.█
```

Time Complexity:

If there are n coin denominations:

- Sorting: $O(n \log n)$
- Checking coins: $O(n)$
- Overall: $O(n \log n)$

If coins are already sorted, it becomes $O(n)$.

Space Complexity:

- Sorting and result storage: $O(n)$
- Space Complexity: $O(n)$